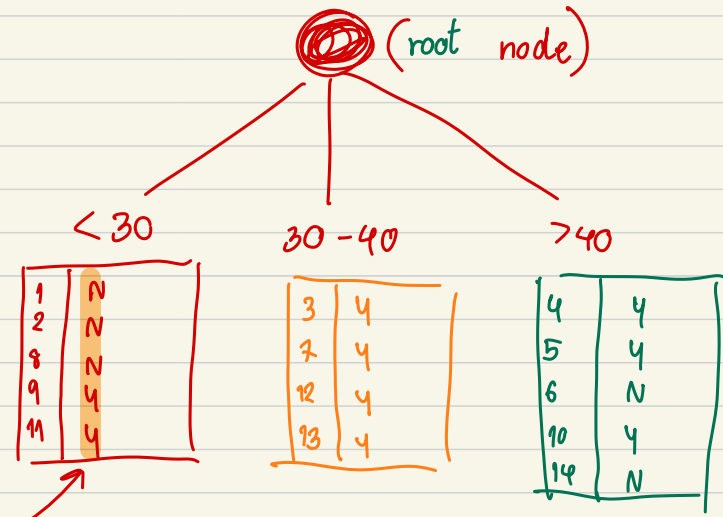


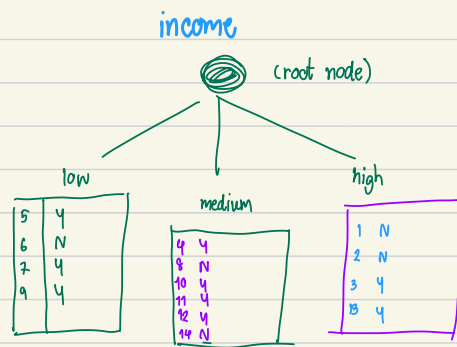
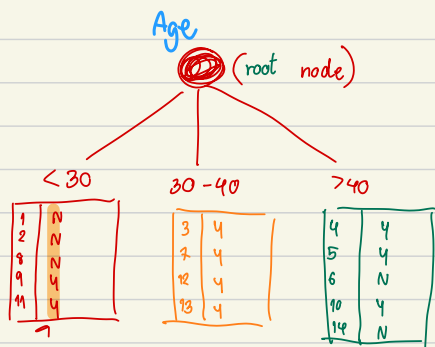
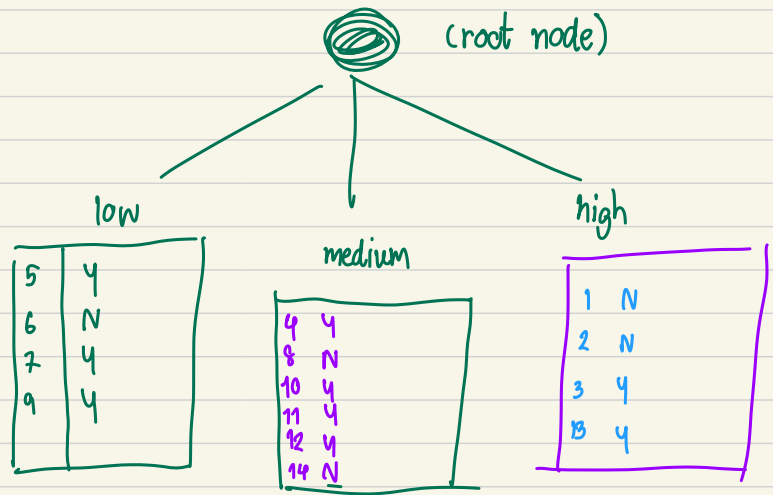
Training data set: Who buys computer?

	age	income	student	credit_rating	buys_computer
1	<=30	high	no	fair	no
2	<=30	high	no	excellent	no
3	31...40	high	no	fair	yes
4	>40	medium	no	fair	yes
5	>40	low	yes	fair	yes
6	>40	low	yes	excellent	no
7	31...40	low	yes	excellent	yes
8	<=30	medium	no	fair	no
9	<=30	low	yes	fair	yes
10	>40	medium	yes	fair	yes
11	<=30	medium	yes	excellent	yes
12	31...40	medium	no	excellent	yes
13	31...40	high	yes	fair	yes
14	>40	medium	no	excellent	no

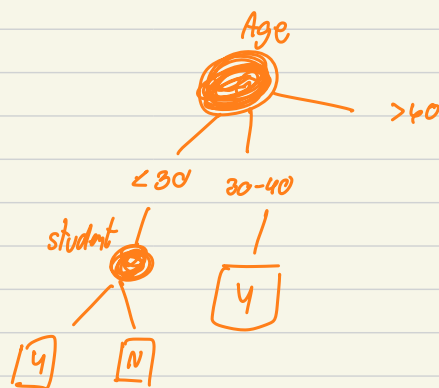
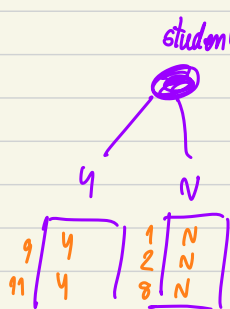
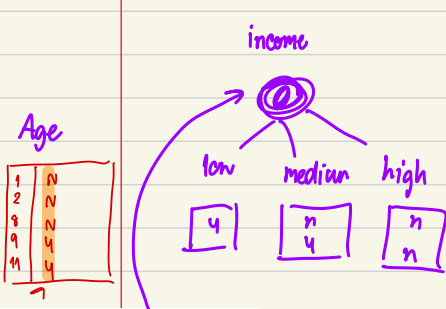


Training data set: Who buys computer?

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3	31...40	high	no	fair	yes
4	>40	medium	no	fair	yes
5	>40	low	yes	fair	yes
6	>40	low	yes	excellent	no
7	31...40	low	yes	excellent	yes
8	<=30	medium	no	fair	no
9	<=30	low	yes	fair	yes
10	>40	medium	yes	fair	yes
11	<=30	medium	yes	excellent	yes
12	31...40	medium	no	excellent	yes
13	31...40	high	yes	fair	yes
14	>40	medium	no	excellent	no

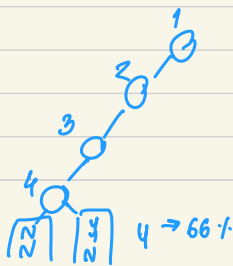


- อันไหนเปลี่ยนดีกว่ากัน
Age เพราะถ้า 30-40 มี 4
ส่วนอันอื่นๆ ก็ดูแบบว่าเปลี่ยนได้
ว่าจริงเป็น 4 หรือ N?
จริงใจหน่อย?



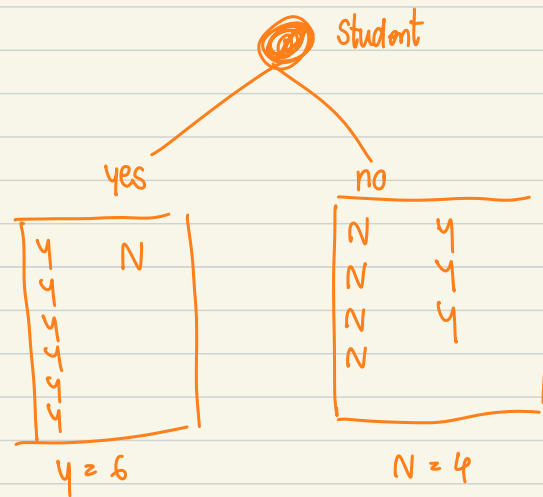
Training data set: Who buys computer?

	age	income	student	credit_rating	buys_computer
1	<=30	high	no	fair	no
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3	31...40	high	no	fair	yes
4	>40	medium	no	fair	yes
5	>40	low	yes	fair	yes
6	>40	low	yes	excellent	no
7	31...40	low	yes	excellent	yes
8	<=30	medium	no	fair	no
9	<=30	low	yes	fair	yes
10	>40	medium	yes	fair	yes
11	<=30	medium	yes	excellent	yes
12	31...40	medium	no	excellent	yes
13	31...40	high	yes	fair	yes
14	>40	medium	no	excellent	no



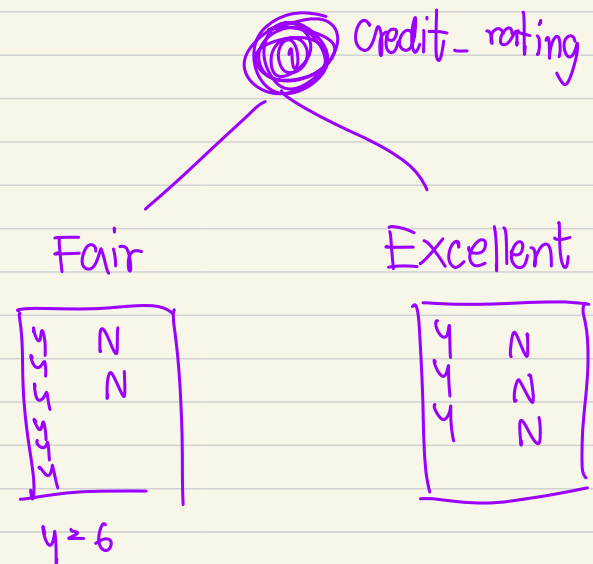
Training data set: Who buys computer?

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no



Training data set: Who buys computer?

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no



HW

Training data set: Who buys computer?

age	income	student	credit_rating	buys computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

1. age

ก่อนจัด - หลังจัด

$$\text{Gain (A)} = \text{Info (D)} - \text{Info}_{\text{age}}(\text{D})$$

หาค่า Info (D)

$$= I(9,5) = -\frac{9}{14} \log_2 \left(\frac{9}{14} \right) - \left(\frac{5}{14} \right) \log_2 \left(\frac{5}{14} \right) = 0.940$$

หาค่า Info_{age}(D)

$$\begin{aligned} & \leq 30 \quad 30-40 \quad > 40 \\ & = \frac{5}{14} I(2,3) + \frac{4}{14} I(4,0) + \frac{5}{14} I(3,2) \\ & = \frac{5}{14} \left[-\frac{2}{5} \log_2 \left(\frac{2}{5} \right) - \frac{3}{5} \log_2 \left(\frac{3}{5} \right) \right] + \frac{4}{14} I(4,0) \\ & \quad + \frac{5}{14} \left[-\frac{3}{5} \log_2 \left(\frac{3}{5} \right) - \frac{2}{5} \log_2 \left(\frac{2}{5} \right) \right] = 0.694 \end{aligned}$$

จะได้ว่า

age	p _i	n _i	I(p _i , n _i)
≤ 30	2	3	0.971
30-40	4	0	0
> 40	3	2	0.971

$$\therefore \text{Gain}_{\text{age}} = 0.940 - 0.694 = 0.246 \#$$

2. Income

$$\text{Gain}_{\text{income}} = \text{Info (D)} - \text{Info}_{\text{in}}(\text{D})$$

หาค่า Info_{in}(D)

$$\begin{aligned} & = \frac{4}{14} I(3,1) + \frac{6}{14} I(4,2) + \frac{4}{14} I(2,2) \\ & = \frac{4}{14} \left[-\frac{3}{4} \log_2 \left(\frac{3}{4} \right) - \frac{1}{4} \log_2 \left(\frac{1}{4} \right) \right] + \frac{6}{14} \left[-\frac{4}{6} \log_2 \left(\frac{4}{6} \right) - \frac{2}{6} \log_2 \left(\frac{2}{6} \right) \right] \\ & \quad + \frac{4}{14} \left[-\frac{2}{4} \log_2 \left(\frac{2}{4} \right) - \frac{2}{4} \log_2 \left(\frac{2}{4} \right) \right] \\ & = 0.232 + 0.394 + 0.285 \\ & = 0.911 \end{aligned}$$

$$\therefore \text{Gain}_{\text{income}} = 0.940 - 0.911 = 0.029 \#$$

3. Student

$$\text{Gain}_{\text{student}} = \text{Info (D)} - \text{Info}_{\text{stu}}(\text{D})$$

หาค่า Info_{stu}(D)

$$\begin{aligned} & = \frac{7}{14} I(6,1) + \frac{7}{14} I(3,4) \\ & = \frac{7}{14} \left[-\frac{6}{7} \log_2 \left(\frac{6}{7} \right) - \frac{1}{7} \log_2 \left(\frac{1}{7} \right) \right] + \frac{7}{14} \left[-\frac{3}{7} \log_2 \left(\frac{3}{7} \right) - \frac{4}{7} \log_2 \left(\frac{4}{7} \right) \right] \\ & = 0.296 + 0.493 \\ & = 0.789 \end{aligned}$$

$$\therefore \text{Gain}_{\text{student}} = 0.940 - 0.789 = 0.151 \#$$

4. Credit_rating

$$\text{Gain}_{\text{credit}} = \text{Info (D)} - \text{Info}_{\text{cre}}(\text{D})$$

หาค่า Info_{cre}(D)

$$\begin{aligned} & = \frac{8}{14} I(6,2) + \frac{6}{14} I(3,3) \\ & = \frac{8}{14} \left[-\frac{6}{8} \log_2 \left(\frac{6}{8} \right) - \frac{2}{8} \log_2 \left(\frac{2}{8} \right) \right] + \frac{6}{14} \left[-\frac{3}{6} \log_2 \left(\frac{3}{6} \right) - \frac{3}{6} \log_2 \left(\frac{3}{6} \right) \right] \\ & = 0.464 + 0.429 \\ & = 0.893 \end{aligned}$$

$$\therefore \text{Gain}_{\text{Credit}} = 0.940 - 0.893 = 0.047 \#$$